BPSC TRE 4.0 • General Science

## TRE's No. 1 Science Topic

# ECOLOGY & ENVIRONMENT

40 of 146 TRE Science Qs

40 Real PYQs | NCERT + SCERT | Outside Facts

Grass ▶ Grasshopper ▶ Frog ▶ Snake ▶ Hawk

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# Where TRE Asked From (Analysis)



## 146 Science questions checked

- 146 science questions from 11 papers (GS part) of TRE 1.0, 2.0, 3.0.
- Most asked: Ecology & Environment = 40 questions (27%)
- Second: Light & the Human Eye (15), then Electricity (8).

## Which book had the answer?

| Source | All 146 | Ecology 40 |
|---|---|---|
| NCERT (6-12) | 74% | 62% |
| Bihar SCERT | 55% | 50% |
| Outside both | 25% | 35% |

## Main NCERT/SCERT chapters

- NCERT 10 Ch13 Our Environment
- NCERT 12 Ch11-13 (Population, Ecosystem, Biodiversity)
- SCERT 10 Ch15 Our Environment, Ch16 Resource Management
- SCERT 8 Ch12, Ch19 Conservation, Pollution

## From outside (Lucent / ecology books)

- 10% law: Lindeman TRE 1.0 (2023)
- Y-shaped model: Odum TRE 1.0 (2023)
- Itai-itai: Cadmium TRE 1.0 (2023)
- Ramsar: Wetlands TRE 1.0 (2023)

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# 1. Components of an Ecosystem

## What is an ecosystem?

- Living + non-living components of an area together = ecosystem (NCERT 10 Ch13).
- Biotic: plants, animals, microbes; Abiotic: temperature, rainfall, soil, minerals.
- Man-made: garden, crop field, aquarium; natural: forest, pond.

## Three roles

- Producers: green plants + some bacteria TRE 1.0 (2023)
- Consumers: herbivores, carnivores, omnivores, parasites
- Decomposers: bacteria + fungi, return nutrients to the soil.

## Exam Trap

★ "Component of an ecosystem: microbes / plants / animals" → all three → (D) More than one TRE 1.0 (2023)

★ Who makes organic from inorganic? Autotrophs + some bacteria → (D)

## Grassland food chain



Grass → Grasshopper → Frog → Snake → Hawk

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# 2. Food Chain + Trophic Levels



## Trophic Levels

| Level | Who | Example |
|---|---|---|
| I | Producers / autotrophs | Grass, trees |
| II | Primary consumers | Deer, goat |
| III | Secondary consumers | Frog |
| IV | Tertiary consumers | Lion, hawk |

## Chains asked in TRE

- Forest: plants → deer → lion
  TRE 1.0 (2023)
- Grass → deer → tiger
  TRE 3.0 (2024)
- NCERT exercise: grass → goat → human
- Branched chain = food web
  TRE 1.0 (2023)

## Trap + Trick

- Insectivorous plants are also producers TRE 3.0 (2024)
- Grass-fish-cow (wrong), grass-rabbit-elephant (wrong)
- Sparrow: eats seeds = I, eats insects = II consumer

## Two types of food chain (NCERT 12)

**Food chain**

**Grazing (GFC)**
Grazing
→ Starts with green plants
Main path in water

**Detritus (DFC)**
Detritus
→ Starts with dead matter
More energy on land

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# 3. Energy Flow: 1% and 10%

## NCERT figures

- Green plants turn ~1% of sunlight into food TRE 1.0 (2023)
- PAR = <50% of incoming radiation; plants capture 2-10% of PAR (NCERT 12)
- 10% law: ~10% energy goes to the next level TRE 1.0 (2023)
- Energy flow is unidirectional; 3-4 links in a chain

## 10% law calculation

| Level | Energy |
|---|---|
| Producers | 10,000 J |
| Primary consumers | 1,000 J |
| Secondary consumers | 100 J |
| Tertiary consumers | 10 J |

## Remember

- Most energy = producers TRE 3.0 (2024)
- GPP − R = NPP
- Trick: 1 - 2 to 10 - 10
- 10% law given by: Lindeman (1942) TRE 1.0 (2023)

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# 4. Ecological Pyramids

## Upright or inverted? (NCERT 12 Ch12)

| Pyramid | Usually | When inverted? |
|---|---|---|
| Number | Upright (grassland) | One big tree + insects + birds |
| Biomass | Upright on land | Sea: less phytoplankton, more fish |
| Energy | Always upright | Never |

## How TRE asked it

★ Which pyramids can be both upright + inverted? Number + biomass → (D) TRE 3.0 (2024)

★ Grassland: ~6 million plants → only 3 top carnivores (pyramid of numbers)

★ Standing crop = living mass at a level at a given time; better measured as dry weight

## Trap

- The energy pyramid is never inverted (heat loss at every level).
- Saprophytes have no place in pyramids.

## Trick

- Energy is always Erect
- Number flips on a Tree, Biomass flips in the Sea

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# 5. Biomagnification + Ozone + Waste



## Biomagnification

- Non-biodegradable pesticides *increase* at every trophic level.
- Highest in humans (at the top)
- Asked twice in TRE 1.0

TRE 1.0 (2023)

- Trap: bioaccumulation = inside one organism; eutrophication = algae from nutrients



## Ozone layer (NCERT 10)

- Ozone (O3) = 3 oxygen atoms; up high it protects from UV
- Falling since 1980s: CFCs (fridges, fire extinguishers)
- 1987: UNEP agreement to limit CFCs
- Ozone at ground level = deadly poison
- UV = high energy, short wavelength TRE 1.0 (2023)
- Ozone: long wavelength / most stable / mesosphere — all three wrong → (E) TRE 3.0 (2024)

## Waste

- Plastic is not broken down by parasites, saprophytes or bacteria → (D) TRE 1.0 (2023)
- Enzymes are specific, so plastic stays.
- Biodegradable: fruit peels, grass, wood, cake
- Kulhads on a large scale = loss of fertile topsoil (NCERT 10)

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# 6. Population + Interactions

## Population (NCERT 12 Ch11)

- Unlimited resources → J-shaped ($dN/dt = rN$)
- Limited resources → S-shaped (logistic)
- Carrying capacity K is set by limited resources TRE 1.0 (2023)

## Interactions

| Interaction | A/B | Example |
|---|---|---|
| Mutualism | +/+ | Lichen, mycorrhiza, fig-wasp |
| Competition | −/− | Flamingo-fish |
| Predation | +/− | Tiger-deer |
| Parasitism | +/− | Cuscuta, koel |
| Commensalism | +/0 | Orchid on mango, egret-cattle |
| Amensalism | −/0 | One is harmed |

## TRE Trap

★ Lichen (alga + fungus) = Mutualism TRE 3.0 (2024)

★ Even if Protocooperation is an option, pick NCERT's Mutualism.

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# 7. Biodiversity + Conservation



## India (NCERT 12 Ch13)

- 2.4% of world land, 8.1% of species
- One of 12 mega-diversity countries
- 34 hotspots; 3 in India: Western Ghats-Sri Lanka, Indo-Burma, Himalaya
- 81,000 fauna + 47,000 flora species TRE 2.0 (2023)

## Evil Quartet

- Habitat loss (biggest cause)
- Over-exploitation
- Alien species invasion
- Co-extinction

## Two ways of conservation

**Conservation**

| In situ (on site) | Ex situ (off site) |
|---|---|
| National parks | Zoos |
| Sanctuaries, biosphere | Botanical gardens |
| Sacred groves | Seed banks, cryo |

## Asked in TRE

★ National park = protects the whole ecosystem TRE 3.0 (2024)

★ Botanical garden is not in situ, it is ex situ TRE 1.0 (2023)

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# 8. Facts from Outside NCERT/SCERT



## Names + models

- 10% law: Lindeman (1942) TRE 1.0 (2023)
- Y-shaped energy model: Odum TRE 1.0 (2023)
- Pioneer on bare rock: Lichen TRE 1.0 (2023)



## Diseases + pollutants

- ★ Itai-itai = Cadmium TRE 1.0 (2023)
- ★ Minamata = mercury; blue baby = nitrate
- ★ SO2 indicator = Lichen TRE 1.0 (2023)
- ★ Secondary pollutants: Ozone, PAN TRE 1.0 (2023)
- ★ Main methane source: paddy fields TRE 1.0 (2023)

## Conservation + places

- Ramsar (1971) = Wetlands TRE 1.0 (2023)
- Himalayan Yew (taxol): Himachal + Arunachal TRE 1.0 (2023)
- Biggest O2 source: oceanic plants TRE 3.0 (2024)
- Breathing roots: halophytes (mangroves) TRE 1.0 (2023)

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# Test Yourself: 9 Questions

**1. About how much of the sunlight falling on leaves do green plants turn into food? (TRE 1.0, 2023)**

(A) 1% : Correct: NCERT 10
(B) 2% : No
(C) 5% : No
(D) More than one : No
(E) None : No
Ans: (A) 1% | ~1% (NCERT 10 Ch13)

**2. How much energy passes from one trophic level to the next? (TRE 1.0, 2023)**

(A) 15% : No
(B) 10% : Correct: 10% law
(C) 5% : No
(D) More than one : No
(E) None : No
Ans: (B) 10% | 10% law

**3. Who gave the 10% law of energy flow in an ecosystem? (TRE 1.0, 2023)**

(A) Elton : Pyramid of numbers
(B) Haeckel : Word 'Ecology'
(C) Lindeman : Correct: 1942
(D) More than one : No
(E) None : No
Ans: (C) Lindeman | Lindeman (1942)

Every option explained + Answer

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# Test Continues: Q4-6

**4. Which of these forms a food chain? (TRE 3.0, 2024)**

(A) Grass, deer, tiger : Correct
(B) Grass, fish, cow : A cow does not eat fish
(C) Grass, rabbit, elephant : Elephant is a herbivore
(D) More than one : Only A
(E) None : A is correct
Ans: (A) Grass, deer, tiger | Grass - deer - tiger

**5. Which pyramids can be both upright and inverted? (TRE 3.0, 2024)**

(A) Number : Yes (tree)
(B) Biomass : Yes (sea)
(C) Energy : Always upright
(D) More than one : Correct: A+B
(E) None : No
Ans: (D) More than one | Number + biomass (NCERT 12)

**6. Rising concentration of toxic substances at successive trophic levels is called (TRE 1.0, 2023)**

(A) Bioaccumulation : In one organism
(B) Biomagnification : Correct
(C) Eutrophication : Algae from nutrients
(D) More than one : No
(E) None : No
Ans: (B) Biomagnification | Biological magnification

Every option explained + Answer

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# Test Continues: Q7-9

**7. Which is NOT included in in situ conservation? (TRE 1.0, 2023)**

(A) Wildlife sanctuary : In situ
(B) National park : In situ
(C) Botanical garden : Correct: ex situ
(D) More than one : No
(E) None : No
Ans: (C) Botanical garden | Botanical garden = ex situ

**8. The relationship between algae and fungi in lichens is called (TRE 3.0, 2024)**

(A) Commensalism : Not +/0
(B) Protocooperation : Not an NCERT term
(C) Mutualism : Correct: +/+
(D) More than one : No
(E) None : No
Ans: (C) Mutualism | Mutualism (NCERT 12)

**9. Itai-itai disease is caused by (TRE 1.0, 2023)**

(A) Cadmium : Correct: Japan
(B) Iron : No
(C) Mercury : Minamata
(D) More than one : No
(E) None : No
Ans: (A) Cadmium | Cadmium

Every option explained + Answer

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# How much do you remember?

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+ 18 Practice MCQs + 15 One-liners

## See you tomorrow at 7 AM:

## BPSC notes

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